# Overview

- Explanation of Plug-in Hybrid Electric Vehicles (PHEV)
- Objectives
- Modeling and Analysis
- Powertrain Architecture and Component Selection
- Operating Strategy and Optimization

# What is a Plug-in Hybrid Electric Vehicle?

## Plug-in HEV Attributes

- Combine function of EVs and HEVs
  - Electric range of 10-60 miles
  - Home recharging at night
  - Continuous driving possible without recharge
- Plug-in HEV uses electricity while battery charge is high
  - Switches automatically to hybrid mode when:
    - Battery charge is low
    - Higher vehicles speeds
    - Driver requires full power

## Keys to Plug-in HEV Efficiency

- Smaller internal combustion engine is more efficient
- Powerful, efficient electric drive provides high performance
- Battery is smaller, lower in cost and mass than a pure EV
- Fuel economy, w/o electricity, equals or exceeds normal HEV.
- ***Displacement of petroleum fuel by electricity significantly increases fuel efficiency.***

# “Big Picture” Objectives

- Energy Security: Reduce transportation fuel demand that now greatly exceeds U.S. fuel supply
- Global Warming: Reduce $CO_2$ emissions
- Environment: Attain air quality targets in critical areas
- Quality of Life: Reduce urban air and noise pollution
- Consumer Values: Increase vehicle efficiency and reliability
  Enhance fueling convenience and choice
  Contribute to national and societal efficiency and environmental goals

# Typical PHEV Powertrain Operation

## Three Operating Modes

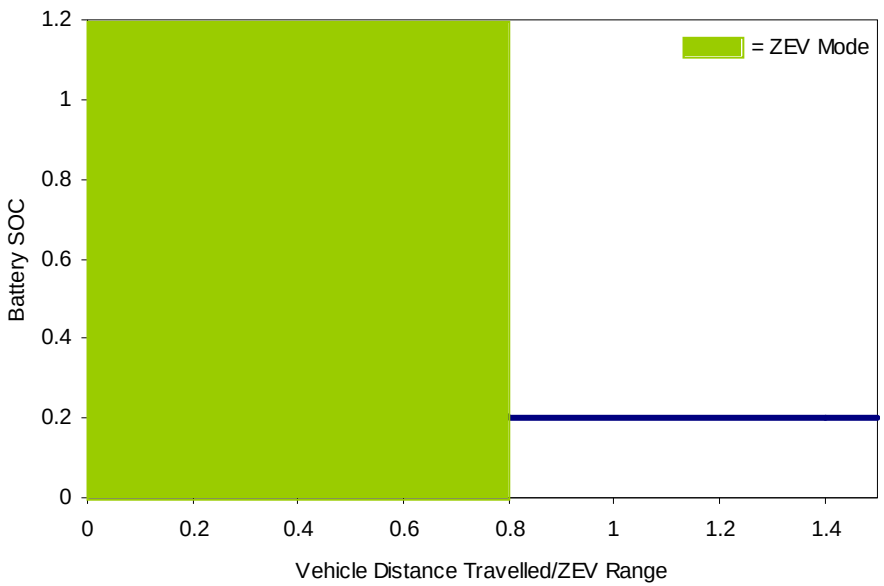
1. Electric Mode – Vehicle operates using only electric motor power provided by battery. Engine is off
2. Charge Depletion Hybrid Mode – Engine is used with E-motor support. Battery is allowed to deplete.
3. Charge Sustain Hybrid Mode – Engine is used with E-motor support. Battery state-of-charge is maintained by charging with excess engine torque

## PHEVs can be

- Series or Parallel Powertrain Configurations
- Fuel cell, gasoline, or diesel hybrids

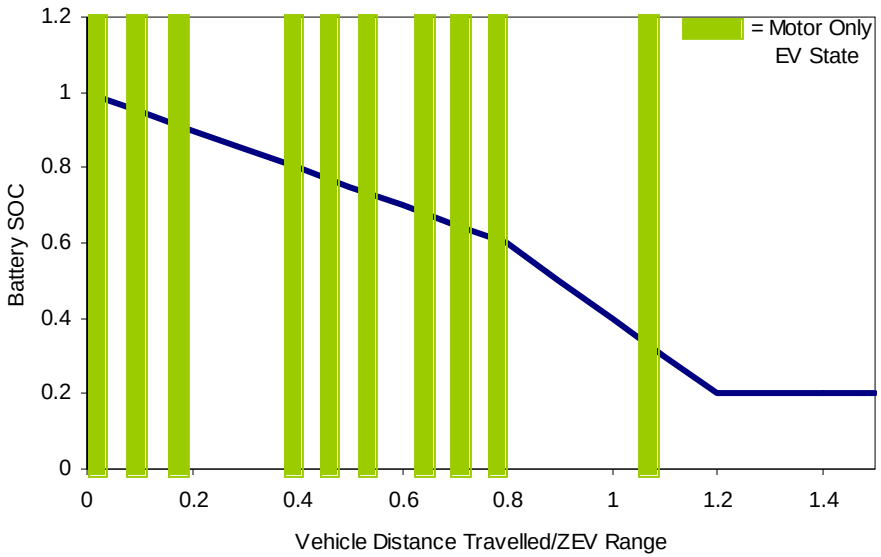
# Range Extending Strategy

- Maximize stored battery energy (to limits of battery performance and durability)
- HEV mode triggers include:
  - Vehicle speed
  - Battery SOC
  - Driver accel request (optional)
- Typically adjust HEV mode speed to accommodate bulk of urban driving
- Higher E/M power requirement
- Charge-sustain HEV operation after EV mode range limit



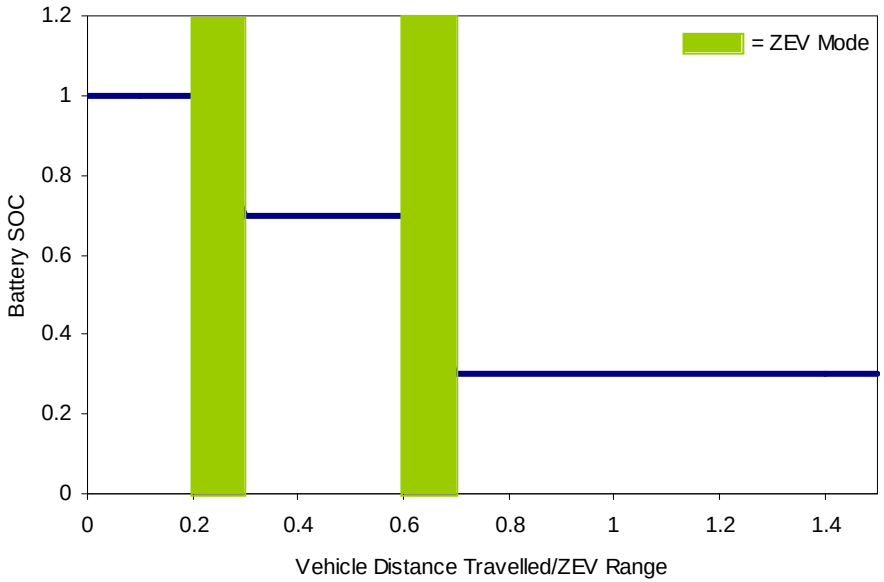
# Charge-Depletion Strategy

- Vehicle is always in HEV mode with regions of electric motor-only operation
- No defined EV operating region (may affect customer acceptance)
- Electric launch
- Lower E/M power requirement
- Charge-sustain HEV mode after minimum battery SOC reached



# “Green Zone” Strategy

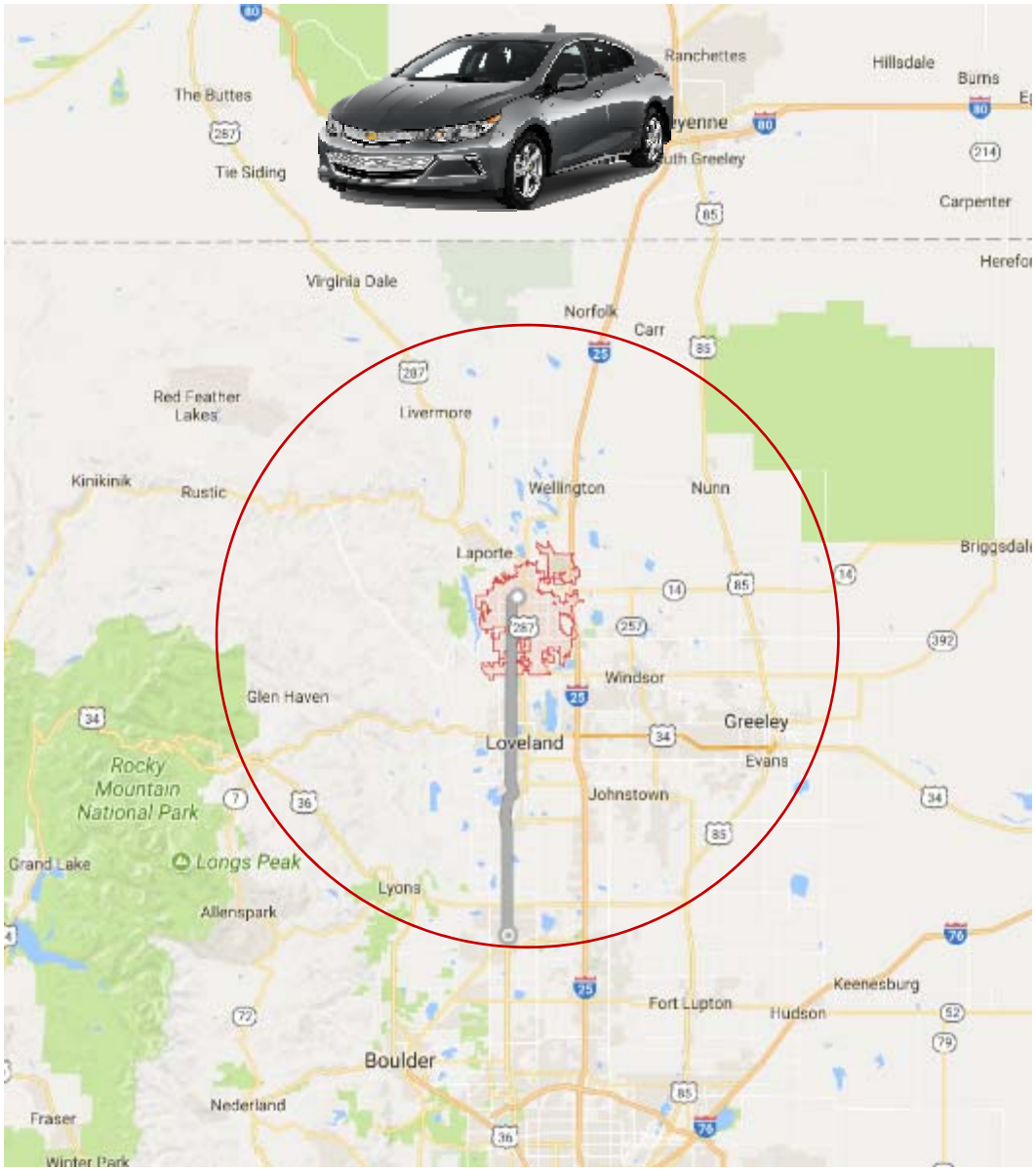
- Geographic regions where EV mode is desirable/required (e.g. inner city or high emissions area)
- Reduced acceleration acceptable and expected
- Trigger by driver or GPS (preferably)
- Challenge to quantify amount of green zone driving
- Need to decide whether to recharge battery to extend green zone driving
- Difficult to maximize battery energy



# How should we evaluate the fuel economy of PHEVs?

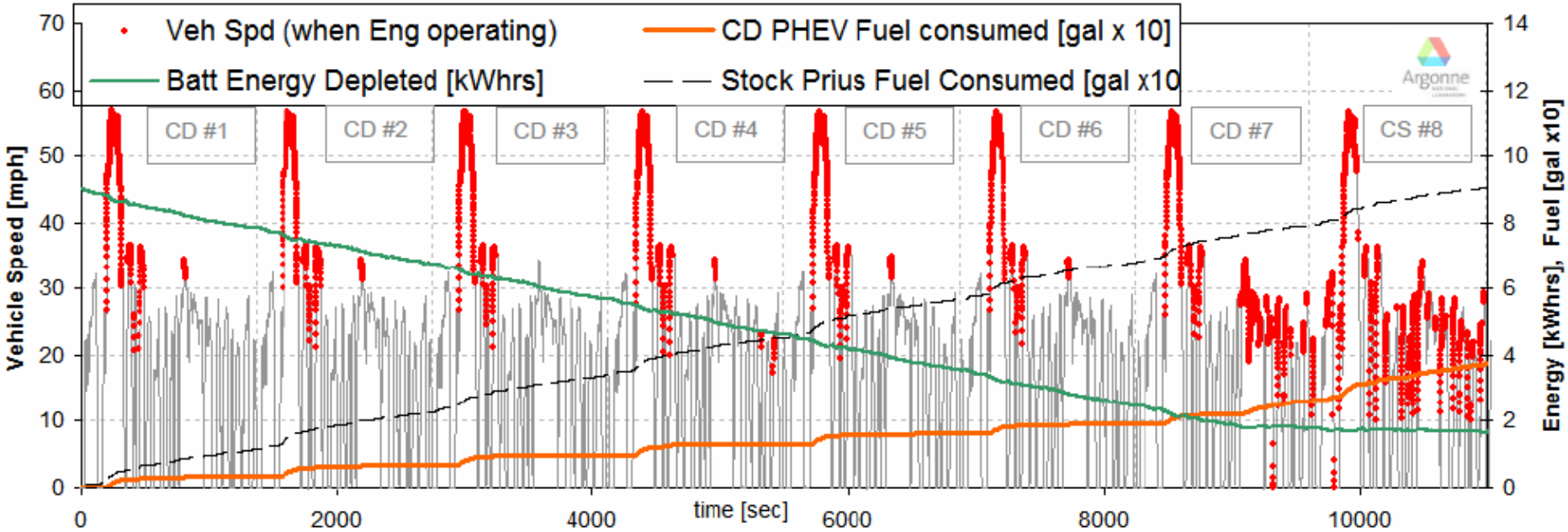
- You can travel anywhere out and back within this circle without using gasoline
  - What is your FE?

- You can go do the airport on a combination of both gasoline and electricity
  - What is your FE?



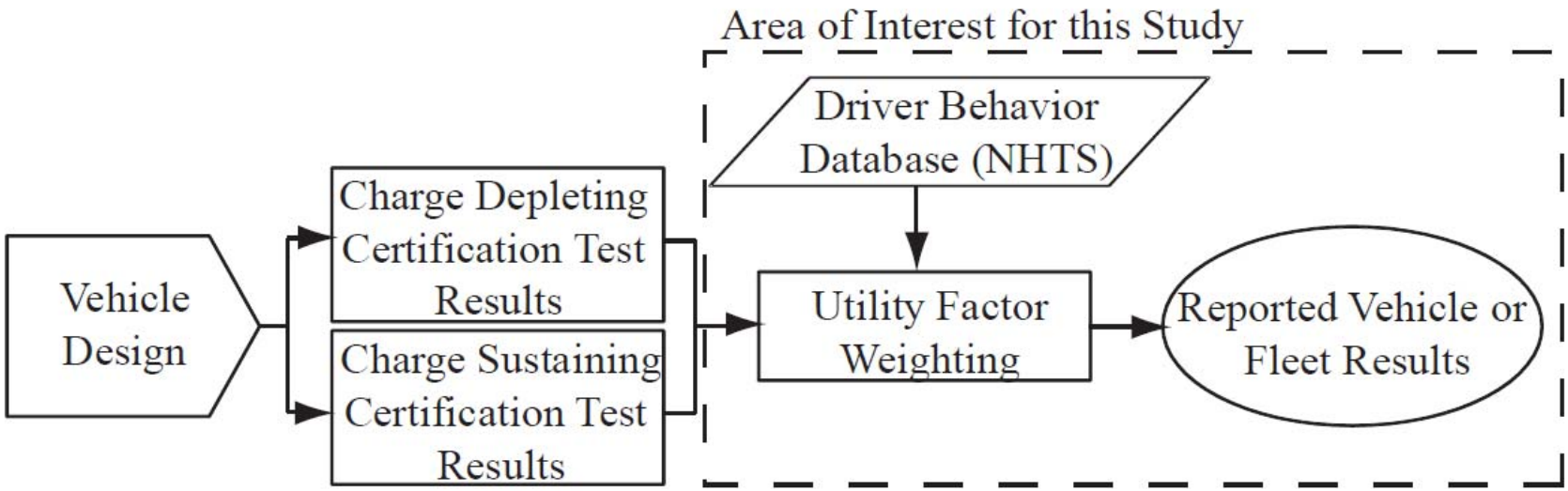
# Introduction

- PHEVs are an important new technology for addressing the sustainability challenges of personal transportation
- Characterizing the performance of PHEVs is challenging
  - Dual fuel (electricity and gasoline)
  - Fuel economy/emissions is distance dependent
  - Fuel economy/emissions is driving behavior dependent



# Introduction, cont.

- To address these challenges, the Society of Automotive Engineers (SAE) has devised the Utility Factor weighted fuel

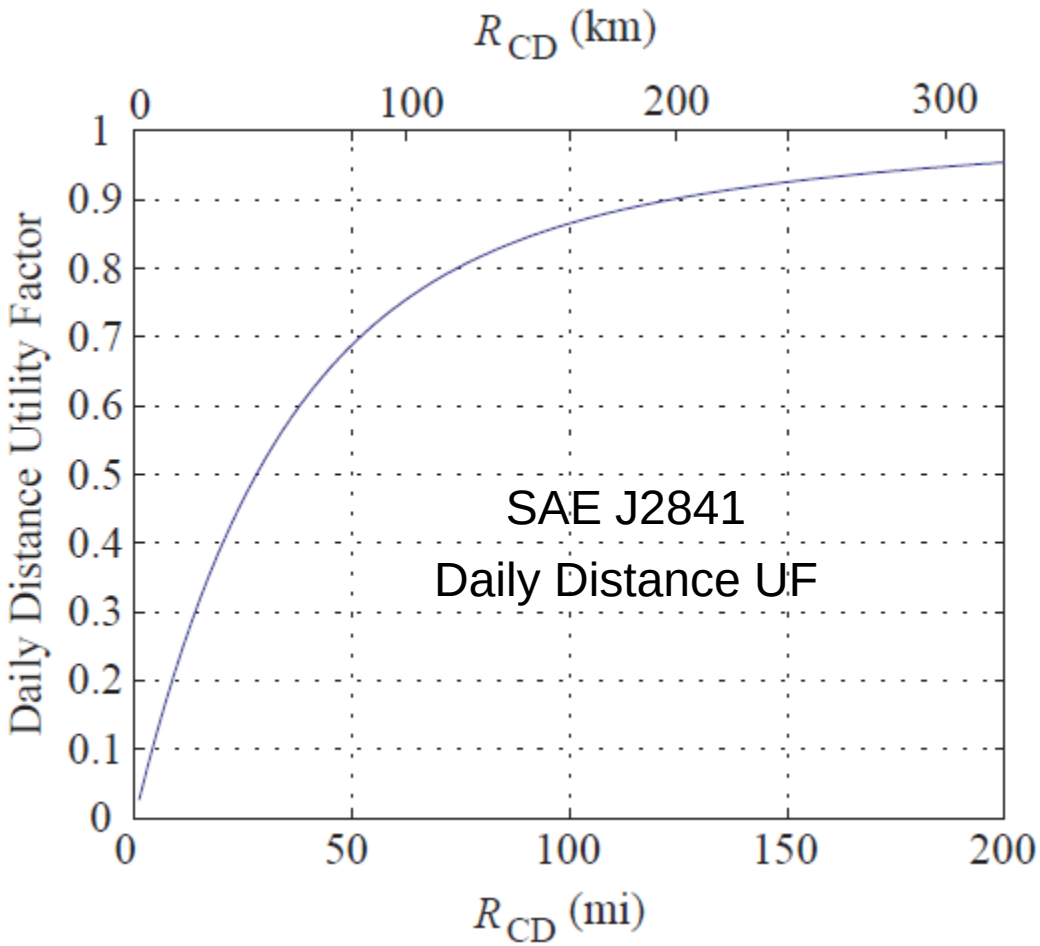



- NHTS is a survey of 69,817 households regarding their driving and travel habits performed every 5-10 years

$$UF_{\text{distance}}\left(R_{CD}\right)=\frac{\sum_{k=1}^{N}\min\left(d(k),R_{CD}\right)}{\sum_{k=1}^{N}d(k)}$$

# Introduction, cont.

- This UF is the basis of MANY Plug-in HEV performance calculations
  - “Sticker” fuel economy
  - EPACT fuel economy (fleets)
  - Corporate average fuel economy (CAFE)
  - California ZEV Mandate (emissions, and credits)
  - Pavley (GHG standards)
  - Low Carbon Fuel Standards
  - Medium and heavy duty FE standards




- How to use the UF? What is the curve ? (Link)

# Using Utility Factor

$$FC_{UF\ weighted} = UF \cdot FC_{CD} + (1 - UF) \cdot FC_{CS}$$

Characteristics

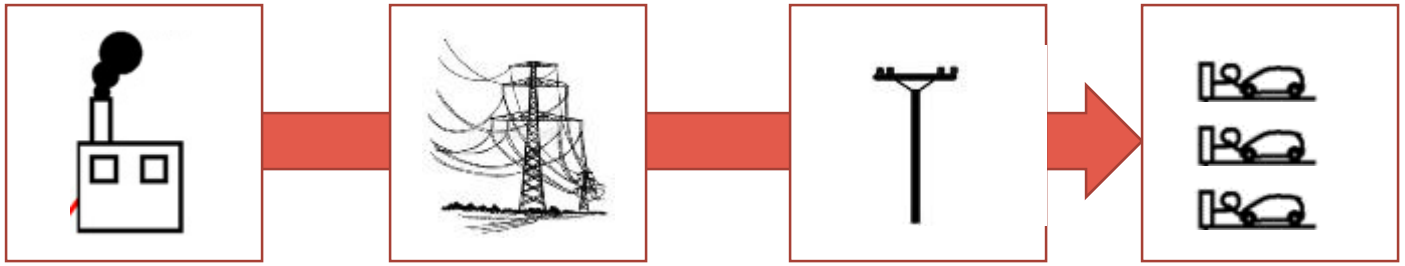
| Vehicle characteristic | Value |
|---|---|
| Charge-sustaining fuel consumption ($FC_{CS}$) | 51.7 mpg (4.55 L (100 km)$^{-1}$) |
| Charge-sustaining electricity consumption ($EC_{CS}$) | 0 |
| Charge-depleting fuel consumption ($FC_{CD}$) | 0 |
| Charge-depleting electricity consumption ($EC_{CD}$) | 282 AC Wh mi$^{-1}$ (175 AC Wh km$^{-1}$) |
| Charge-depleting range ($R_{CD}$) | 42 mi (68 km) |
| Battery energy capacity | 10 DC kWh requiring 11.75 AC kWh to charge |

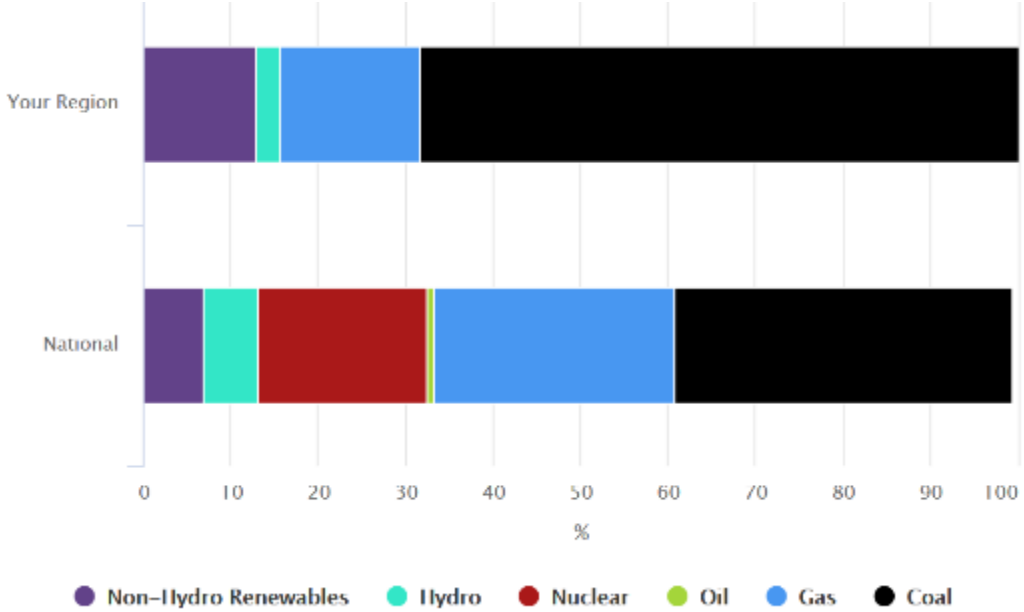
- $R_{CD}$ = 42mi , on the UF graph, UF = 0.635

  $FC_{UFWeighted}$ = 0.635*0 + (0.365)*4.55 = 1.66L/100km

  $FE_{UFWeighted}$= 141.7 mpg

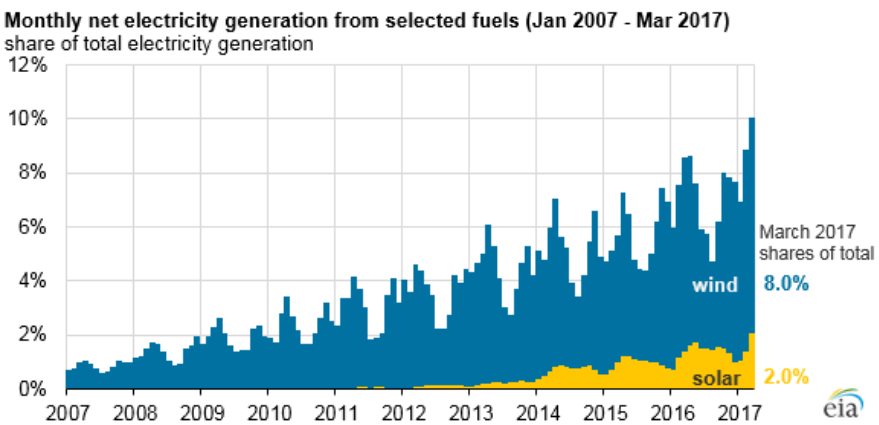
# Electricity Sources

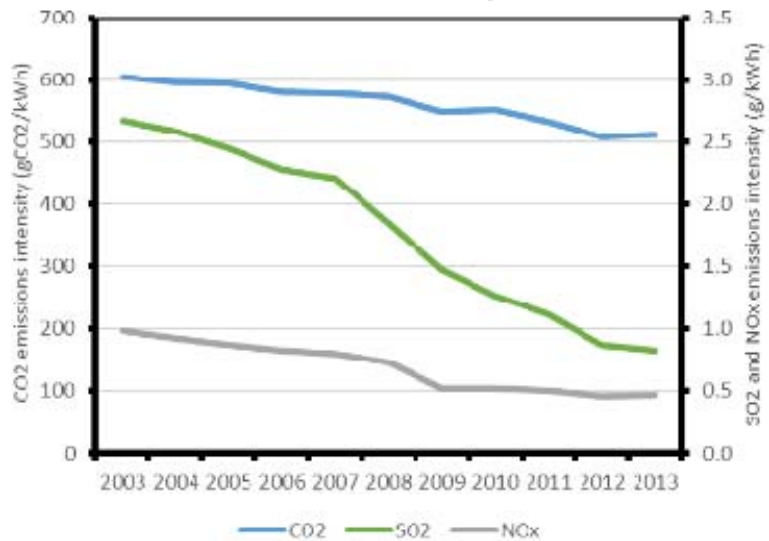




# Electrification of Transportation

- The advantages of electrification come from
  - Fuel switching (centralization of emissions)




20M tons SOx in 1980
4M tons SOx in 2012

  - Vehicle Engineering
    - Hybridization, fewer engine startups, charge-depleting range, home refueling, regenerative braking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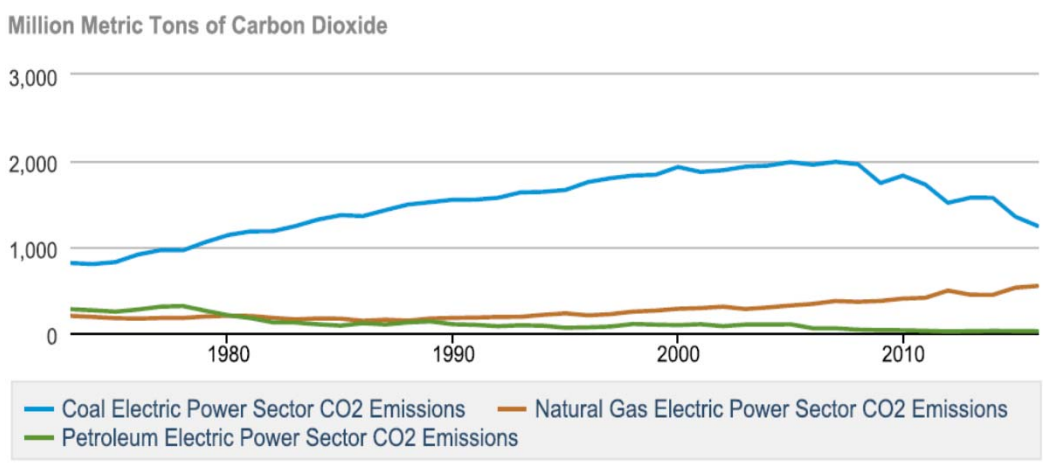
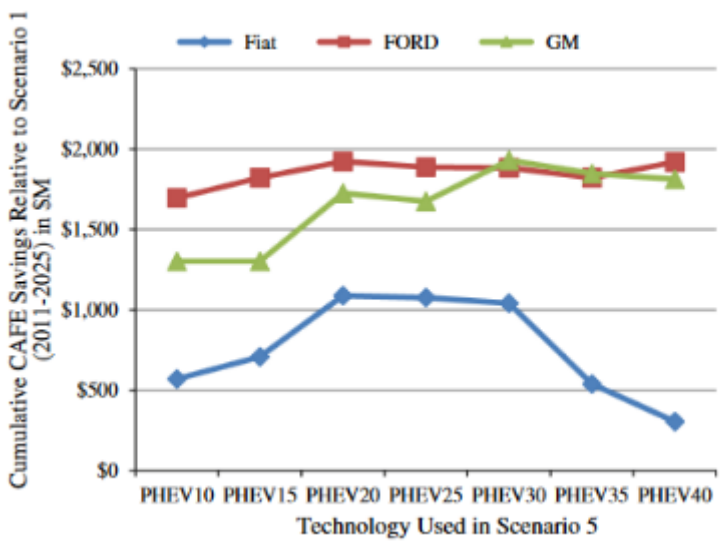
# Electrification of Transportation

- Chevrolet Volt use 26% of the gasoline of a 40 mpg car
- Sustainability of PEVs is improving every day
- There are strong policy incentives for production and sales of PEVs
  - CAFE and CA ZEV Program




eia Source: U.S. Energy Information Administration



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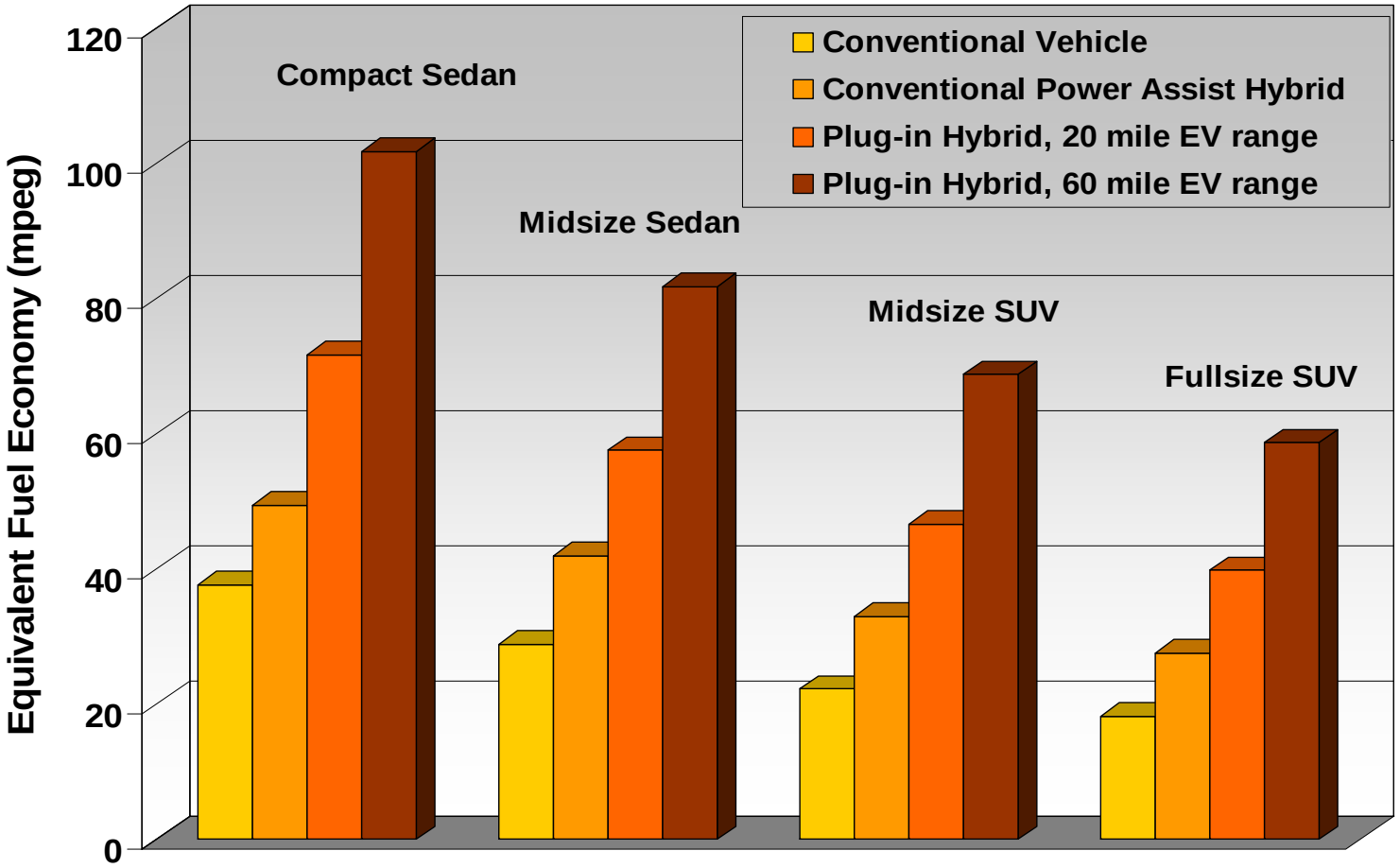
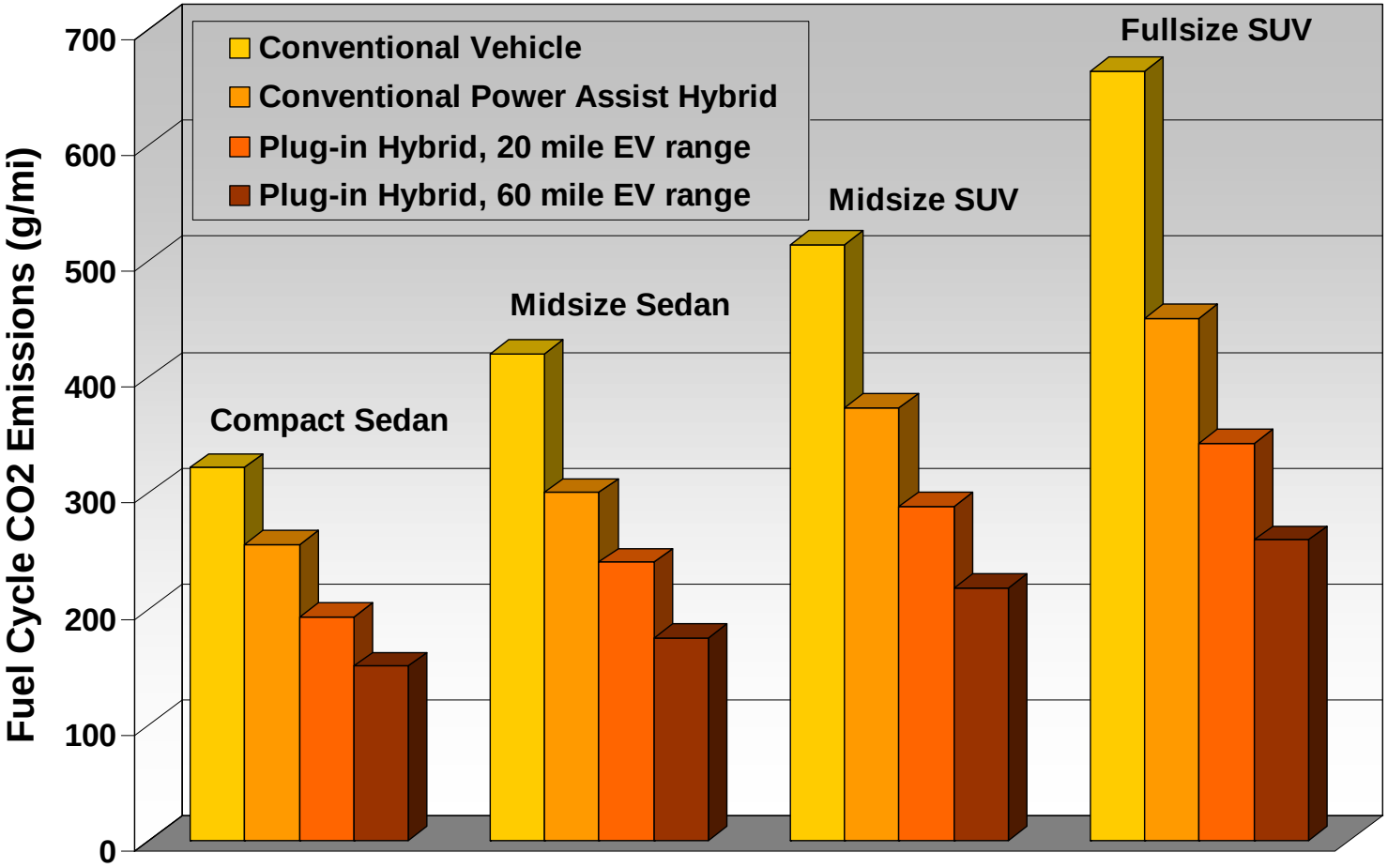
# Equivalent Fuel Economy

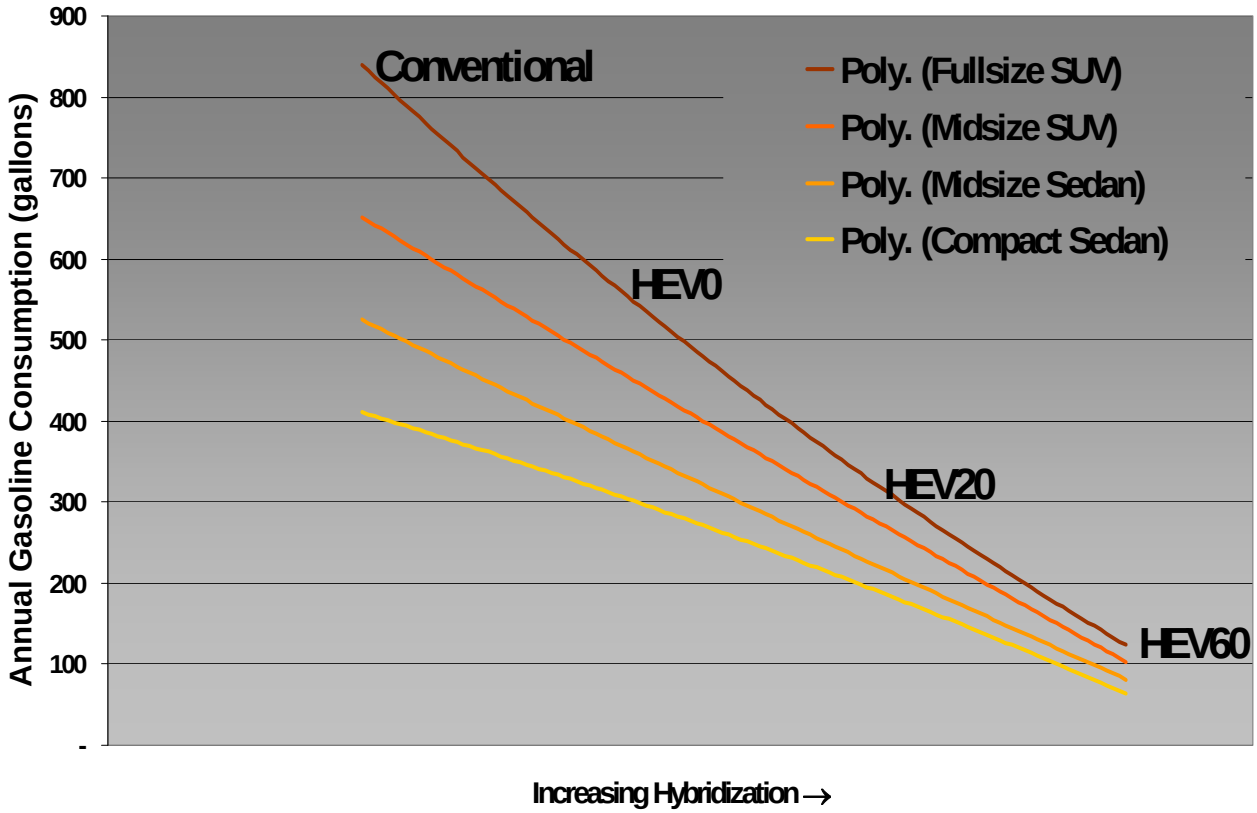


# Greenhouse Gas Emissions

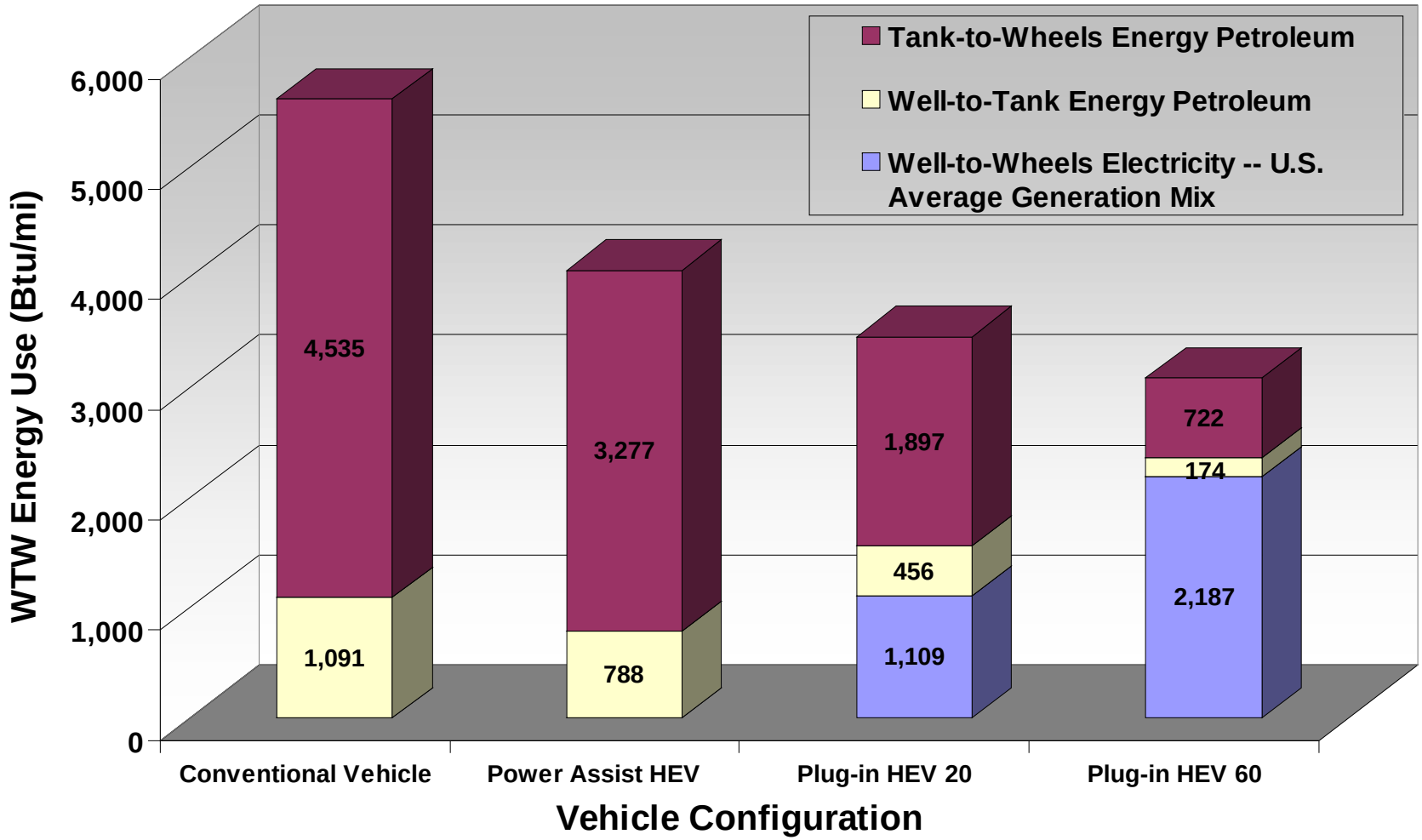


# Fuel Consumption

## Annual Gasoline Consumption



# Well-to-Wheels Energy Use
## Midsize Sedan



# Is Plugging In a Good Customer Attribute?

**CUSTOMER PREFERENCE:**
**Gasoline Station Visit vs. Plugging in At Home**

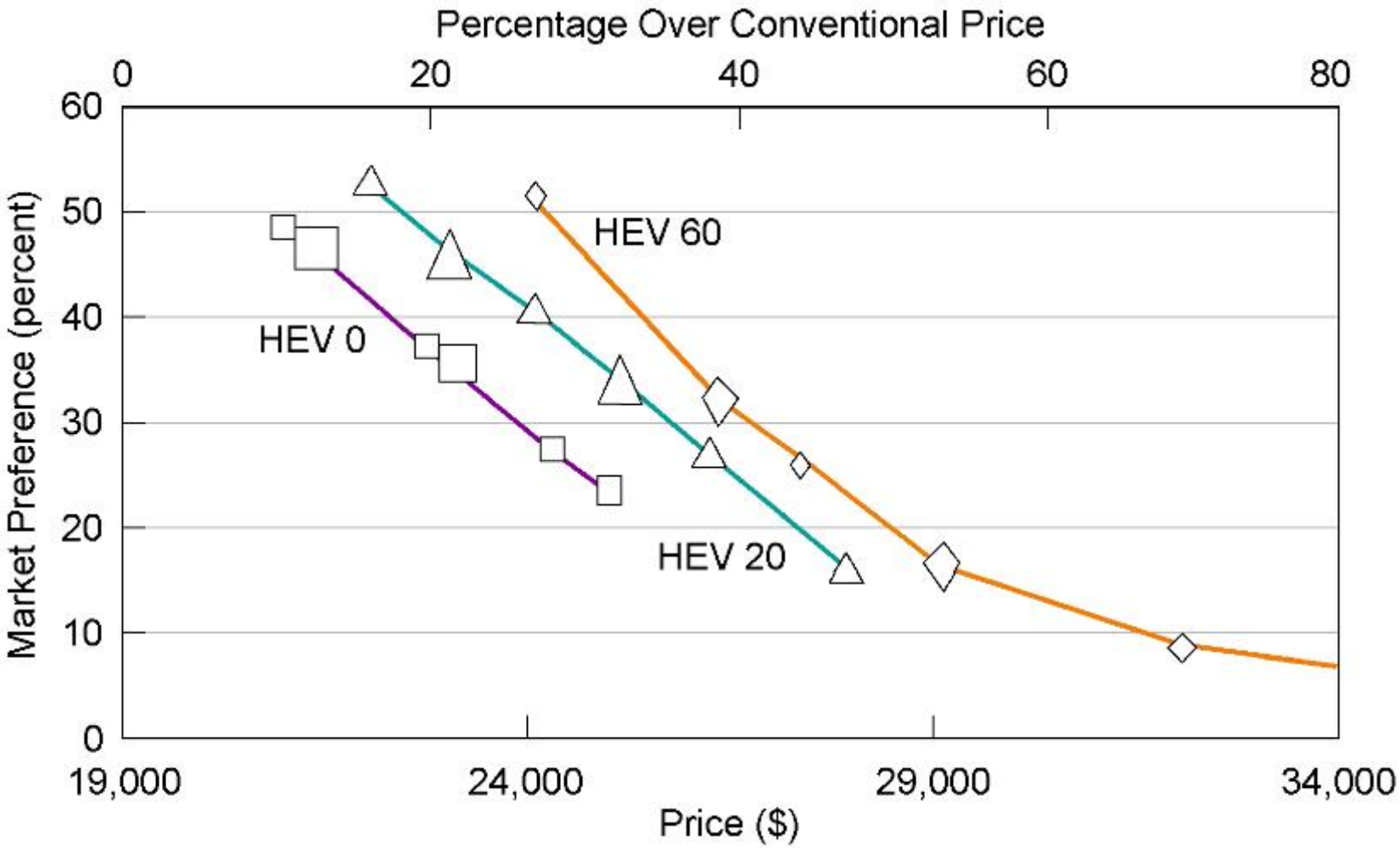
**Ranking Scale**:

- 1 = strongly prefer to fuel vehicle with gas at the gas station
- 9 = strongly prefer to fuel vehicle by plugging it in at home

**Ranking Results:**

- 1-3 Ranking 1.1%
- 4-6 Ranking 35.9%
- 7-9 Ranking 63.0%

# HEV Market Potential vs. Price



# Component Selection

- Battery (must now be sized for CD Range)
  - EV 1 = 115 DC*Wh/mile*
  - Midsized car (Tesla Model S) = 344 DCWh/mile

- Electric Motor (E-motor) (should it maintain performance under both CD and CS modes?)
- Electric Accessories (must work under both CD and CS modes)
  - HVAC System
  - DC-DC Converter
  - Electric brake assist, Electric cooling fans, Electric coolant pumps, Electric power steering
  - Battery charger

# EV, PHEV and HEV BATTERY REQUIREMENTS[1]

| Vehicle Type | Specific Energy (Wh/kg) | Specific Power (W/kg) | Deep Cycle Life (80%△SOC) | Shallow Cycle Life (+/- 50Wh) | Specific Cost ($/kWh) |
|---|---|---|---|---|---|
| Electric Vehicle (EV) | >100-120 | >300-400 | >1000-1500 | ~same as MHEV | <100-150 |
| **Plug-In HEV (PHEV)** | **>60-30[3]** | **>400-800[3]** | **>2000-3000[3]** | **~same as MHEV** | **<200-400[3]** |
| Power-Assist HEV (HEV) | >30-40 | >800 | NA | >300,000 | <500-1000 |

[1] **For vehicles of comparable weight, performance, and cost**

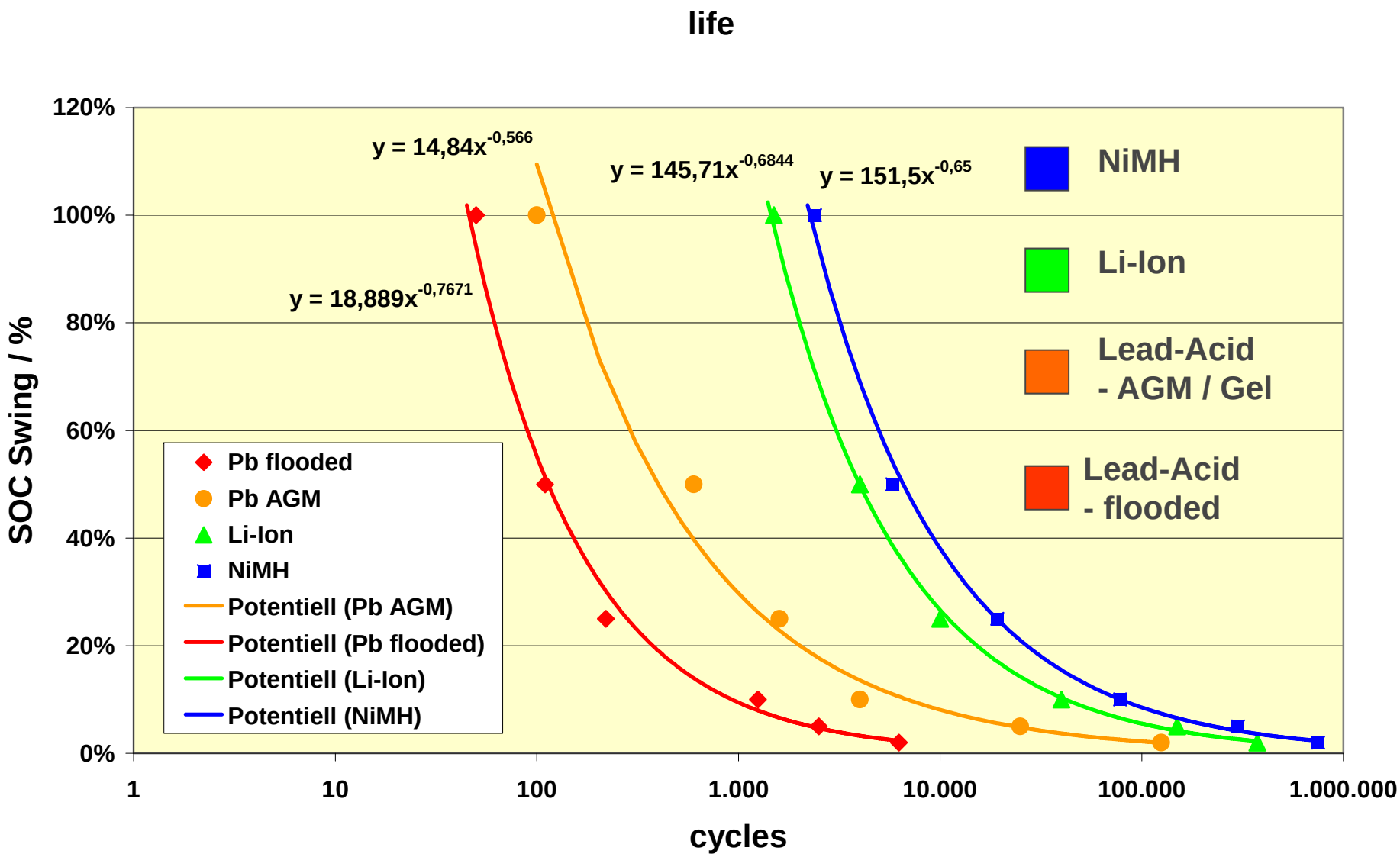
[2] **Vehicle "electric" efficiency of 4 miles/kWh, ~25% better than state-of-the-art HEVs**

[3] **First number in range applies to PHEV-60, second to PHEV-20**

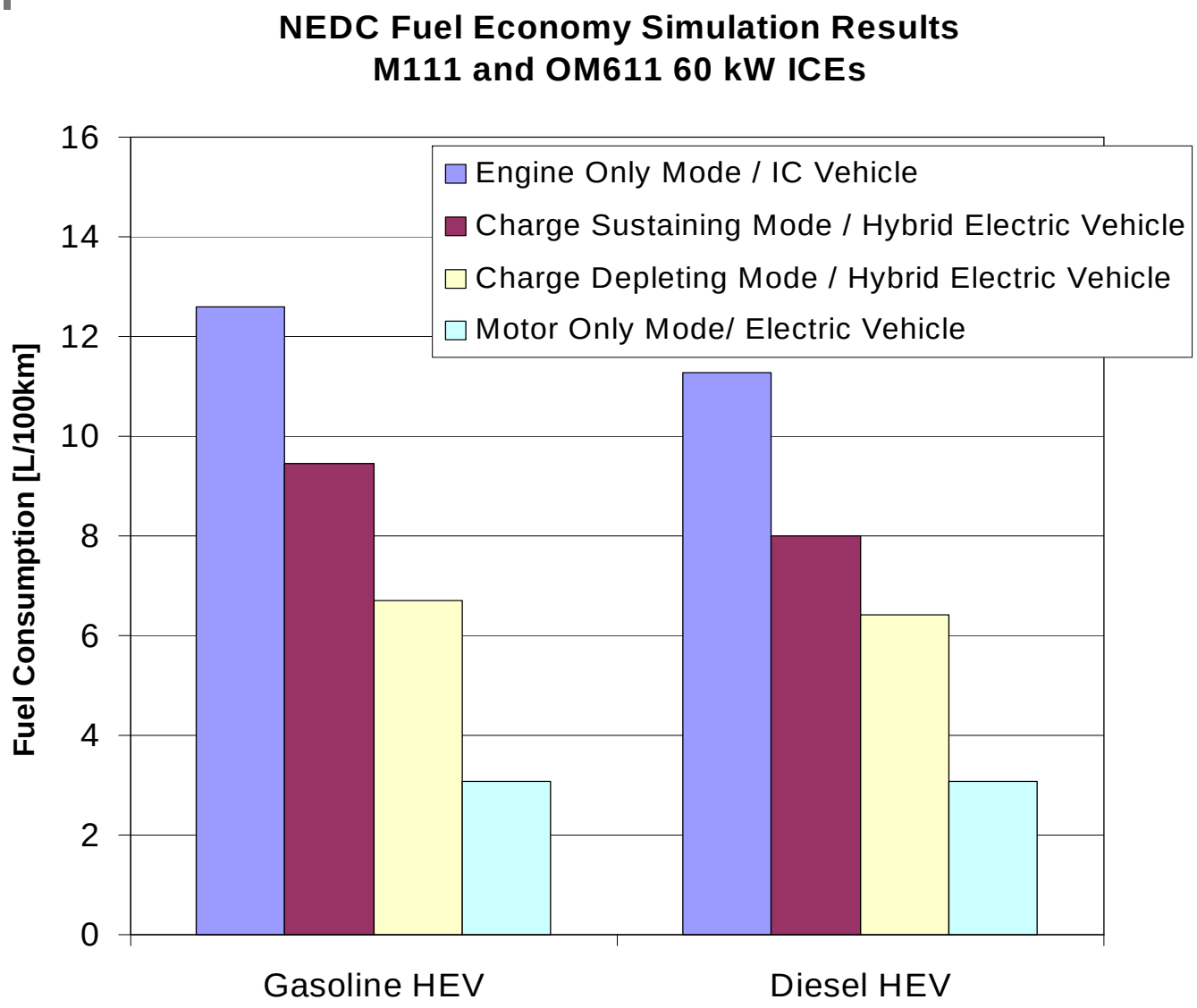
# Comparison of Deep Cycle Batteries PbA, NiMH, Li-Ion

| High Energy Design in Deep Cycle Application | | AGM (PbA) | Li Ion |
|---|---|---|---|
| **energy density** | [Wh/kg] | 35 | > 140 |
| **power output** | [W/kg] | 150 | 800-1500 |
| **charge acceptance** | [W/kg] | 50 | >600 |
| **life time - number of cycles** | at 80% swing | 125 | 2.500 |
| | at 5% swing | 50.000 | 140.000 |
| **cost level** | US-D / kWh | 150 | <150 |

# Cycle Life of Different Battery Chemistries



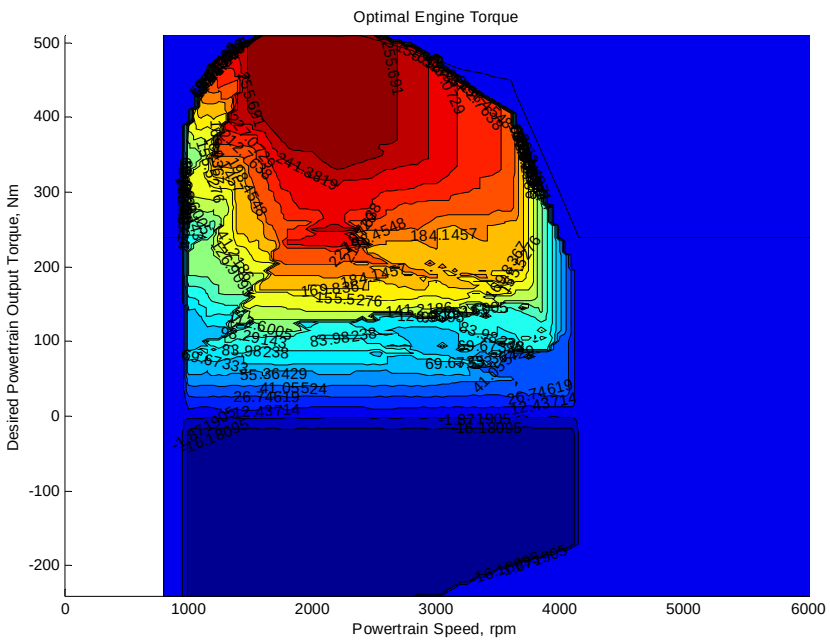
# Vehicle Modeling: Preliminary Results



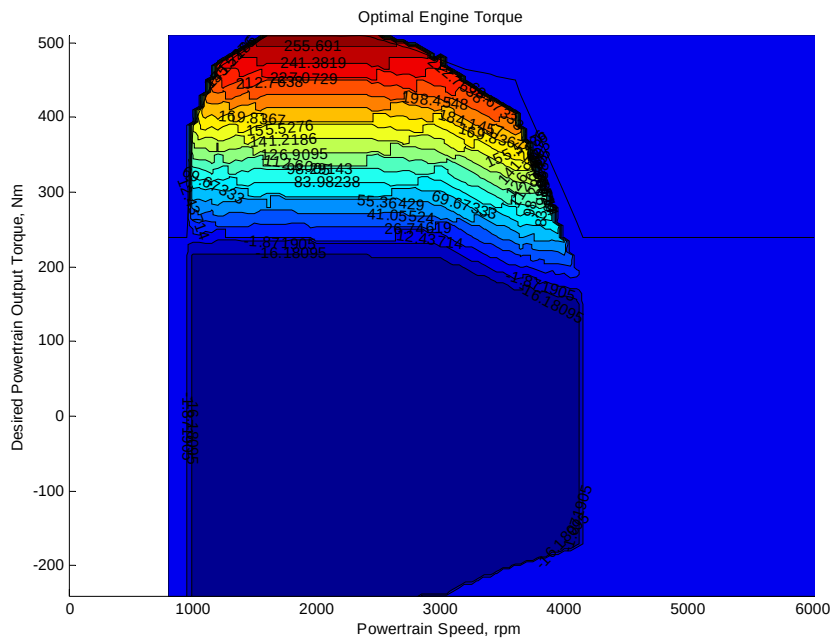
# Plug-In HEV Operating Strategy

# Vehicle Fuel Economy – Torque Control

- Results of Optimization
  - *T_engine = f(rpm, T_command)*



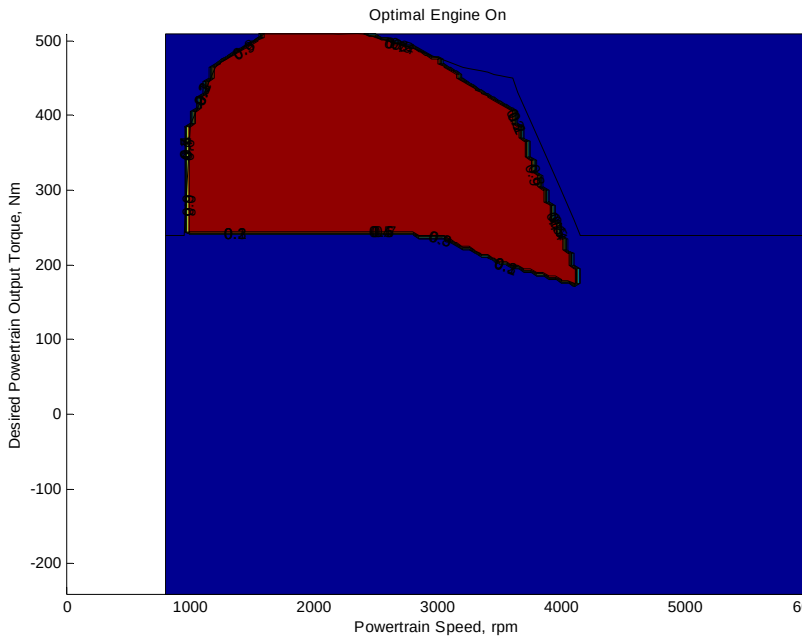

*Battery Sustaining*



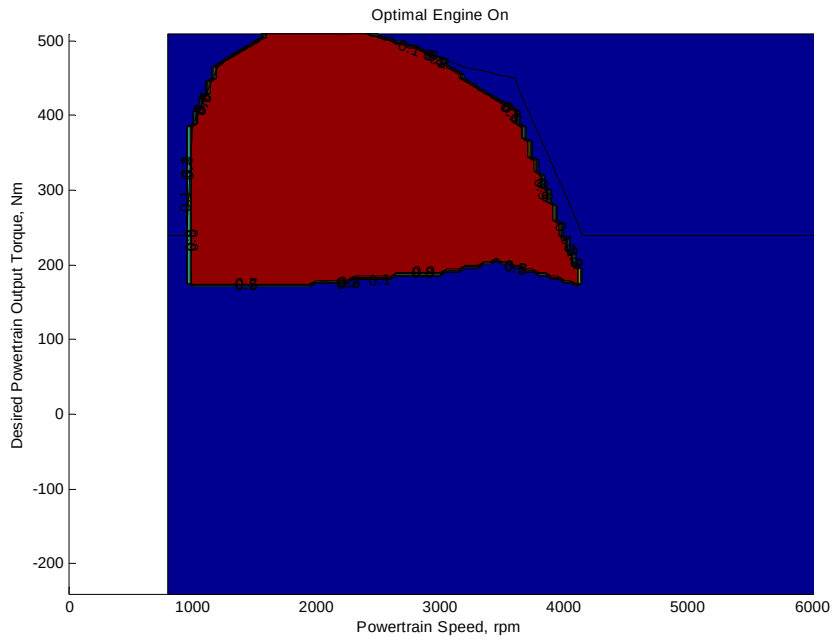

*Battery Depleting – Fueling Cost Optimized*

# Vehicle Fuel Economy – Engine On/Off

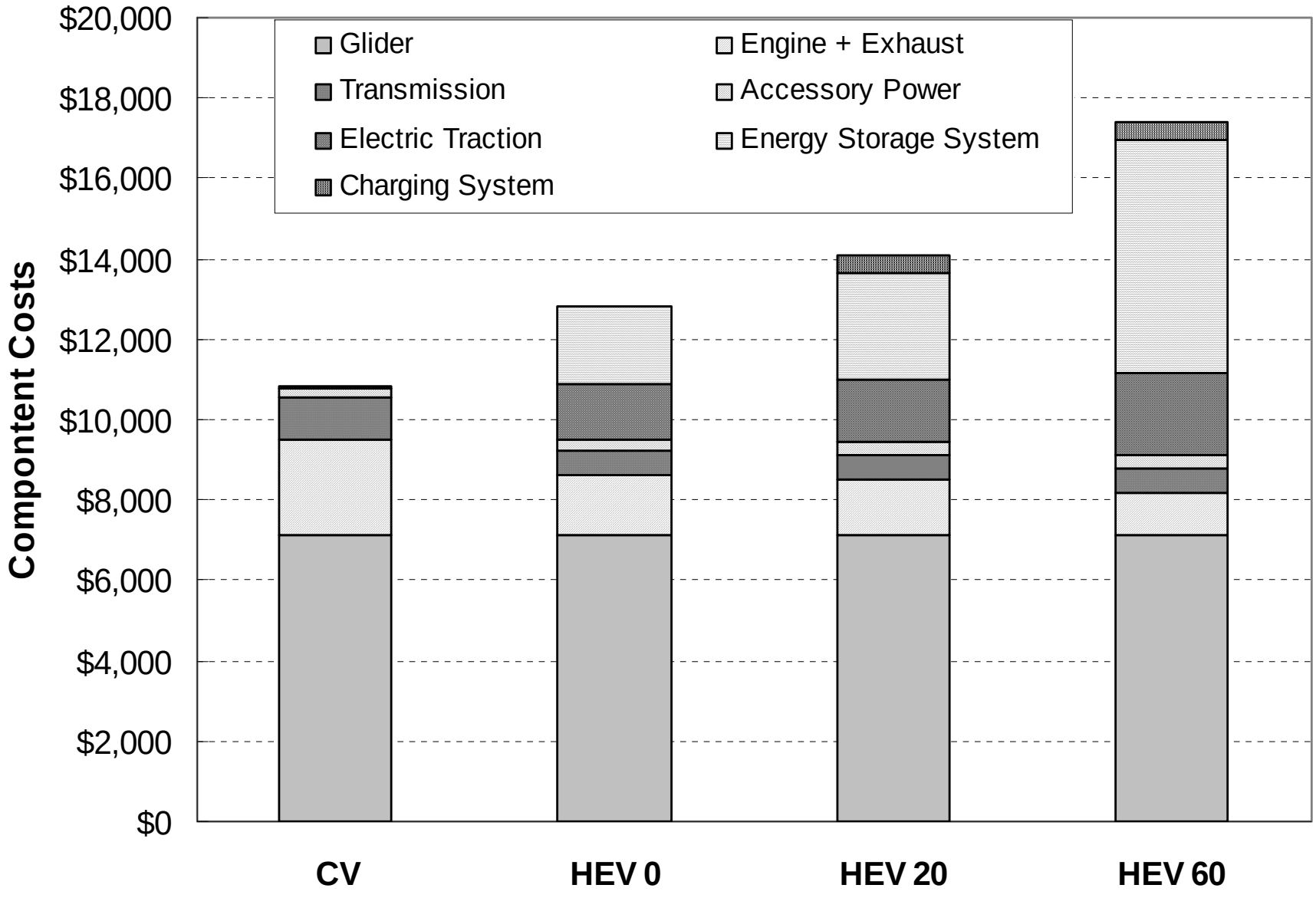
- Using a similar evaluation technique one can consider whether the engine should be on.
  - *Engine_On_Cmd = f(rpm, T_command)*




*Battery Depleting – Fueling Cost Optimized*




*Battery Sustaining*

- For Engine On/Off Control there are other considerations/compromises to be made

# VEHICLE COST COMPARISON